

# Python Notes

## Print statement

```
print("hello world!") # this is for printng hello
```

## Comments

```
'''  
analjnakj  
asjbakjsbda  
asbkjasbckja  
'''  
# jvnsj kv
```

## data type

```
name = 'ganesh '  
age = 24.2
```

```
type(name)
```

```
type(age)
```

```
name = "John"  
age = 25  
height = 5.9
```

```
type(name)  
type()
```

```
type(age)
```

```
type(height)
```

## input and output

```
print("hi")
```

```
input(" what is your age ?")
```

```
input("what is your name ?")
```

```
# cost = int(input("enter the cost ?"))  
name = input('your name ! ')
```

```
type(name)  
# int(input("ascas")) -> integer
```

## how to define a variable

```
name = 'a'  
NaMe = "b"
```

```
name1 = 'a'
```

```
name_1 ="a"
```

```
name1 = "a"
```

```
name = input("what is your name ? ")  
print(name, "hi!" )  
age = int(input("what is your age ?"))  
print("you are", age,"yrs old")
```

```
age = int(input("what is your age ?"))
```

```
'''  
print()  
data type - int, float, str  
input() - int(input("basn")) , float(input("anas"))  
type()  
variables - define
```

```
name = "amirtha ganesh "
```

```
print(f'hi {name}. How are you ?')
```

```
arithmetic operations
if else
boolean data type
print
small project

election day
```

## DAY - 2

```
name = "John" # variable 1
age = 20 # variable2
```

```
name = "ganesh"
```

```
# print(age)
# print(name , age)
# print("Hello " +name)
# print("Hello", name, "you are ", age,"years old")

print(f" hello {name} you are {age} years old. ")

# print("hello {} you are {} years old.".format(name,age))

# {}, .format(var1, var2)

# print("My name is {} and I am {} years old.".format(name, age))
```

```
# Normal print
print("Hello")

# Print multiple values
print(name, age)

# Using +
print("Hello " + name)

# Using comma
print("Hello", name)

# Using f-string
print(f"My name is {name} and I am {age} years old.")

# Using format()
print("My name is {} and I am {} years old.".format(name, age))
```

```
# name = "Alice"
# age = 22
# height = 5.8
# salary = 50000
```

```
# print(name)
# print(age)
# print(height)
# print(salary)
```

```
# 4. Boolean Data Type
```

```
5>10
```

```
# print(10 == 10)
# print(10 != 5)
```

```
# 100 /10
10 /3
# Floor Division
5/2
5 % 2
```

```
# 4 ** 3 # -> 4 power 3
```

```
4* 4*4
```

```
# 5. Arithmetic Operators
```

```
a = 10
b = 3
```

```
print("Addition :", a + b)
print("Subtraction :", a - b)
print("Multiplication :", a * b)
print("Division :", a / b)
print("Floor Division :", a // b)
print("Modulus :", a % b)
print("Power :", a ** b)
```

```
# 6. if - else Statement
```

```
age = int(input("enter the age: "))
```

```
# > - greater than
# < less than
# == equal to
# != not equal to
```

```
if age >= 18:
    print("you are eligible to vote")
else:
    print("you can't vote")
```

```
if condition:
```

```
body
elif condition:
    body
elif conditon:
    body
else:
    body
```

```
A grade > 90
B grade > 75
c grade > 50
else : fail
```

```
# 8. if elif else

marks = int(input("enter your mark ."))

if 100 <= marks >= 90:
    print("Grade A")
elif 89 <= marks >= 75:
    print("Grade B")
elif 74 <= marks >= 50:
    print("Grade C")
else:
    print("Fail")
```

## Day 3

```
print("python" ,"java" , sep = " + ")
```

```
# Using end
print("Python", end = " ")
print("Programming", end = " ")
print("question")
```

```
a = b = c = 18

print(a)
print(b)
print(c)
```

```
# Using sep
print("Python", "Java", "C++", sep=" | ")

# Using end
print("Python", end=" ")
print("Programming")
```

```

# Type Casting

# number = "100"
# type(number)

# 100 -> int -> int(variable_name)
# 100 -> float -> float(variable_name)

# # number = int(number)
# type(number)
# print(float(number))

num = 100

print(str(num))

```

```

a = 10
b = 20
print(a == b)
print(a != b)
print(a > b)
print(a < b)
print(a >= b)
print(a <= b)

```

```

# Nested if

age = 21
citizen = "no"

if age >= 18:
    if citizen == "yes":
        print("Eligible to Vote")
    else:
        print("Not an Indian Citizen")
else:
    print("Under Age")

```

```

# =====
# 1. INTRODUCTION
# =====

print("Hello, World!")

# =====
# 2. OUTPUT
# =====

name = "John"
age = 20

# Normal print

```

```

print("Hello World")

# Print multiple values
print(name, age)

# Using +
print("Hello " + name)

# Using comma
print("Hello", name)

# Using f-string
print(f"Hello {name}, you are {age} years old.")

# Using format()
print("Hello {}, you are {} years old.".format(name, age))

print("-----")

# =====
# 3. VARIABLES
# =====

# Variable Assignment
name = "Alice"
age = 22
salary = 50000
type()
print(name)
print(age)
print(salary)

# Multiple Assignment
a = b = c = 100

print(a)
print(b)
print(c)

# Multiple values in one line
x, y, z = 10, 20, 30

print(x)
print(y)
print(z)

# =====
# 4. DATA TYPES
# =====

integer_value = 100
float_value = 99.99
string_value = "Python"
boolean_value = True
none_value = None

print(integer_value)
print(float_value)

```

```

print(string_value)
print(boolean_value)
print(none_value)

# =====
# 5. INPUT & TYPE CASTING
# =====

name = input("Enter your name : ")

age = int(input("Enter your age : "))

height = float(input("Enter your height : "))

print(name)
print(age)
print(height)

# Type Casting

number = "100"

print(int(number))
print(float(number))

num = 200

print(str(num))

print("-----")

# =====
# 6. OPERATORS
# =====

a = 20
b = 3

# Arithmetic Operators
print("Addition :", a + b)
print("Subtraction :", a - b)
print("Multiplication :", a * b)
print("Division :", a / b)
print("Floor Division :", a // b)
print("Modulus :", a % b)
print("Power :", a ** b)

prin

print()

# Comparison Operators
print(a == b)
print(a != b)
print(a > b)
print(a < b)
print(a >= b)
print(a <= b)

```



```

# Bitwise Operators
print(5 & 3)
print(5 | 3)
print(5 ^ 3)
print(~5)
print(5 << 1)
print(5 >> 1)

print("-----")

# =====
# 7. CONDITIONAL STATEMENTS
# =====

# if

age = 20

if age >= 18:
    print("Eligible to Vote")

print()

# if - else

age = 15

if age >= 18:
    print("Eligible")
else:
    print("Not Eligible")

print()

# if - elif - else

marks = 82

if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 50:
    print("Grade C")
else:
    print("Fail")

print()

# Nested if

age = 21
citizen = "yes"

if age >= 18:
    if citizen == "yes":
        print("Eligible to Vote")
    else:
        print("Not an Indian Citizen")
else:
    print("Under Age")

print()

```

# Problem Statement - 1: Student Grade Calculator

## Objective

Write a Python program that calculates the **total marks**, **average marks**, and **grade** of a student.

## User Input

Your program should ask the user to enter:

- Student Name
- Marks in Subject 1
- Marks in Subject 2
- Marks in Subject 3

## Tasks

- Calculate the **Total Marks**
- Calculate the **Average Marks**
- Display the student's **Grade**

## Grade Criteria

| Average Marks | Grade |
|---------------|-------|
| 90 - 100      | A     |
| 75 - 89       | B     |
| 50 - 74       | C     |
| Below 50      | Fail  |

## Expected Output

The program should display:

- Student Name
- Total Marks
- Average Marks
- Grade

```
student_name = input("enter your name : ")
s1 = float(input("enter marks in subject 1 : "))
s2 = float(input("enter marks in subject 2 : "))
s3 = float(input("enter marks in subject 3 : "))

total_mark = s1 + s2 + s3
total_mark

avg_mark = total_mark/3
avg_mark

print(f" Student name : {student_name}")
print(f" Total Marks : {total_mark}")
```

```

print(f" Average Marks : {avg_mark}")

print("Average Marks :", avg_marks )

if 100 >= avg_mark >=90:
    if 100 >= avg_mark >= 95:
        print("your Grade is S")
    else:
        print("your grade is A")
elif 89 >= avg_mark >= 75:
    print("Grade B ")
elif 75 >= avg_mark >=50:
    print("Grade C ")
else:
    print("Fail")

```

avg\_mark

## Problem Statement - 2: Movie Ticket Booking System

### Objective

Write a Python program that calculates the total ticket price based on the customer's age and the number of tickets.

### User Input

Your program should ask the user to enter:

- Customer Name
- Age
- Number of Tickets

### Ticket Pricing Rules

| Age                | Ticket Price |
|--------------------|--------------|
| Below 5 years      | Free         |
| 5 - 17 years       | ₹150         |
| 18 - 59 years      | ₹250         |
| 60 years and above | ₹180         |

### Tasks

- Determine the ticket price based on the customer's age.
- Calculate the total amount to be paid.
- Display the customer details and the total ticket cost.

### Expected Output

The program should display:

- Customer Name
- Age
- Ticket Price

- Number of Tickets
- Total Amount